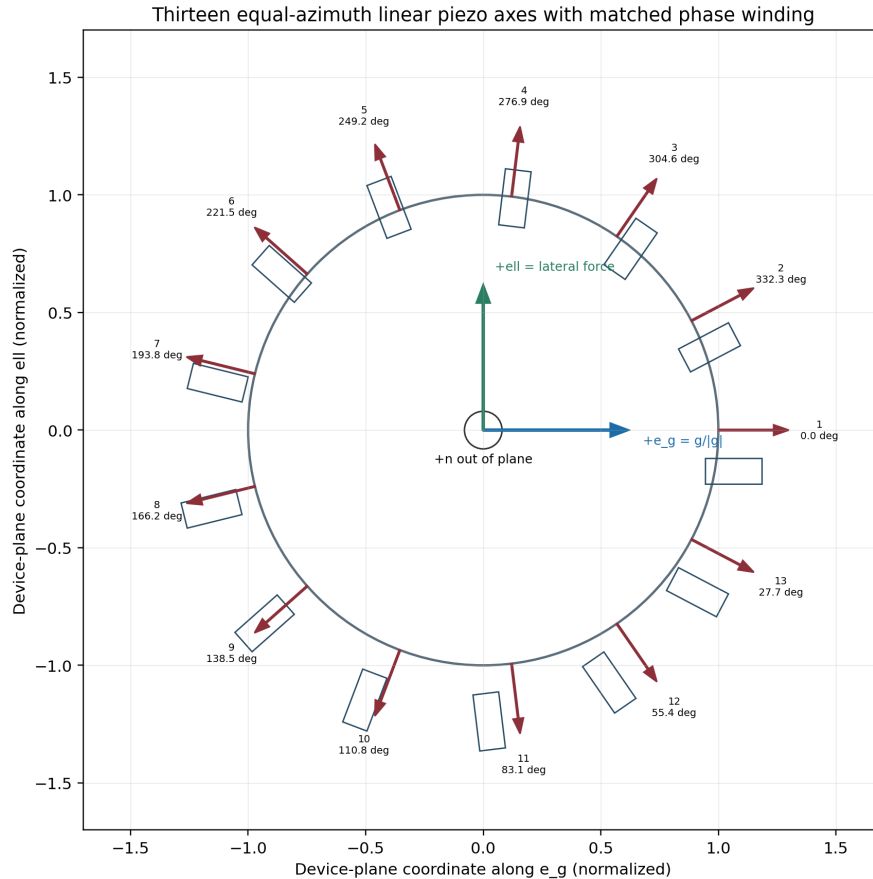


K-Field 13-Element Thruster

Optimized coherent 13-phase linear-piezo array for lateral residual coupling during Earth-CMB traversal



Frontispiece. The active source plane contains thirteen equally spaced linear piezo axes. The electrical phase winding is locked to the mechanical azimuth so that the normalized coherent modal coordinate executes an exact circle in the optimized plane.

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Document Contract and Abstract

Field	Value
Document class	Archival foundational education reference
Coordinate frame	ICRS Cartesian for field vectors; nanometres for Cell3 physical geometry; local orthonormal device coordinates for the piezo plane
Principal investigator	P. Dwayne Esterline, Krytronx, LLC
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Numerical source policy	All numerical values are carried from the supplied machine-readable records or are deterministic evaluations of equations and explicit design constraints stated in this document.
Force-status policy	The source rotating-tube reduction is treated as the calibrated K-Field law. High-frequency and 13-element values are equation evaluations with the transfer factor H retained explicitly; they are not represented as completed force measurements.
PDF generation	Python ReportLab; built-in core fonts only; dynamic table widths; wrapped Paragraph objects in every text cell; repeated table headers.
Companion artifact	K-Field_13_element_thruster_1785003663.json

This paper derives and specifies a planar array of thirteen individually linear piezoelectric oscillators whose axes and carrier phases synthesize one coherent circular modal coordinate. The construction extends the rotating-tube K-Field reduction without requiring any individual proof mass to execute a rigid circular orbit. The array is placed in the physical plane spanned by the normalized response vector $\mathbf{e}_g = \mathbf{g}/|\mathbf{g}|$ and the optimized lateral direction $\mathbf{ell}$. Its oriented plane normal $\mathbf{n}_{opt}$ is selected so that $\mathbf{n}_{opt}$ is perpendicular to $\mathbf{g}$ and the resulting force $\mathbf{n}_{opt} \times \mathbf{g}$ is also perpendicular to the Earth-CMB velocity. The exact ICRS normal is $(-0.315865036407, 0.646162683024, -0.694768354089)$, corresponding to right ascension 116.050867377 degrees and declination -44.008760899 degrees. The positive lateral direction is $(0.077469585422, 0.747375863362, 0.659869519070)$.

Each oscillator axis lies in the device plane at azimuth $\theta_j = 2\pi(j-1)/13$ and is driven with phase $\phi_j = \phi_0 - \theta_j$. The thirteenth roots of unity remove the common normal component and produce two equal, orthogonal quadratures. After normalization by $2/13$, the coherent coordinate is $A[\cos(\omega t + \phi_0)\mathbf{e}_g + \sin(\omega t + \phi_0)\mathbf{ell}]$, an exact circle with unit mechanical circularity. The source law then gives $F_K = M_{active} \pi \kappa_F f C_{pol} H (\mathbf{n} \times \mathbf{g})$. At maximum orientation, $\pi \kappa_F |\mathbf{g}| = 8.989366088879e-5 \text{ N/(kg Hz)}$.

The intrinsic frequency objective is to maximize $f \eta_m(f) H(f)$, where η_m is active modal participation and H is the unresolved coherence, localization, material, wave-vector, temperature, and boundary transfer. Geometry alone does not create a finite upper optimum. Under the selected 50-100 MHz implementation window, the nominal design frequency is therefore 100 MHz. The carrier period is 10 ns, the adjacent phase delay is 0.769230769 ns, and Earth advances 3.698125020 mm in the fixed CMB frame during one carrier cycle. At 100 MHz, the ideal maximum acceleration is $8989.366088879 \text{ H m/s}^2$ and the ideal force is 0.898936609 H N for 100 mg of coherently active mass.

Reading and Status Conventions

Source-exact identity. A value or equation copied directly from the supplied Cell3, rotating-tube, linear-inertia, or circular-oscillator records.

Derived identity. An algebraic or numerical consequence of the source-exact objects. The optimized plane, element vectors, phases, delays, and frequency ledger are in this class.

Engineering continuation. A design constraint or transfer model added to connect the source law to a realizable resonator. The selected 50-100 MHz window, 100 MHz nominal carrier, active modal participation η_m , and coherent transfer H_{13} are engineering continuations.

Measurement statement. A value obtained by experiment. This report contains no newly reported force measurement; all force numbers are source-law evaluations and retain H explicitly.

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1. Device Objective and Complete Optimized Result

The device is a single planar source module containing thirteen equal linear piezoelectric elements. The moving coordinate of each element is one-dimensional, but the phase-ordered displacement field is two-dimensional. The design objective is to maximize the K-Field lateral residual associated with the coherent oriented loop while retaining an electronically reversible force direction and a geometry that can be mechanically balanced.

The source rotating-tube law depends on the oriented local path and on the Earth-CMB response vector $g = G_{\text{phase}} V_E$. A rigid rotor is one implementation of the path. The thirteen-element construction replaces rigid circulation with a spatially distributed traveling phase pattern. The relevant coherent order parameter is the vector sum of the thirteen linear coordinates after the exact 2/13 normalization derived in Section 7.

The optimized plane normal is perpendicular to g , which maximizes $|n \times g|$. The remaining one-parameter degeneracy is removed by requiring the resulting force to be purely lateral to the Earth-CMB velocity. This selects n_{opt} as the normalized component of V_E perpendicular to g . The device plane is then $\text{span}(e_g, ell)$, where $e_g = g/|g|$ and $ell = n_{\text{opt}} \times e_g$.

All thirteen axes use the same maximum available tilt from the plane normal. With no fabrication constraint, the optimum is $\alpha = 90$ degrees, so every axis lies in the plane. Equal azimuth spacing and the matched phase law $\phi_j = \phi_0 - \theta_j$ attain the maximum circular coherent amplitude permitted by thirteen unit-stroke linear actuators. Reversing the sign of the phase winding reverses the oriented loop and the lateral residual without changing the hardware orientation.

Quantity	Optimized value	Role
Element count	13	Uniform odd phase ring; exact thirteenth-root closure
Plane normal n_{opt}	(-0.315865036407, 0.646162683024, -0.694768354089)	Maximum magnitude and purely lateral force selection
Plane normal RA/Dec	116.050867377 deg / -44.008760899 deg	ICRS installation direction
Positive lateral direction ell	(0.077469585422, 0.747375863362, 0.65986951907)	Predicted force direction for negative phase winding
Axis tilt α	90.000000 deg from plane normal	Maximum projected stroke and oriented area
Axis azimuths	$\theta_j = 360 \text{ deg } (j-1)/13$	Equal angular coverage
Drive phases	$\phi_j = \phi_0 - \theta_j$	Positive chirality and + ell force
Nominal carrier	100.000000 MHz	Upper endpoint of selected 50-100 MHz design window
Adjacent delay	0.769230769 ns	T/13 at 100 MHz
Ideal coefficient	8.989366088879e-05 N/(kg Hz)	Maximum orientation, unit circularity, unit transfer
Ideal 100 MHz acceleration	8989.366088879 H m/s^2	Equation evaluation; H retained

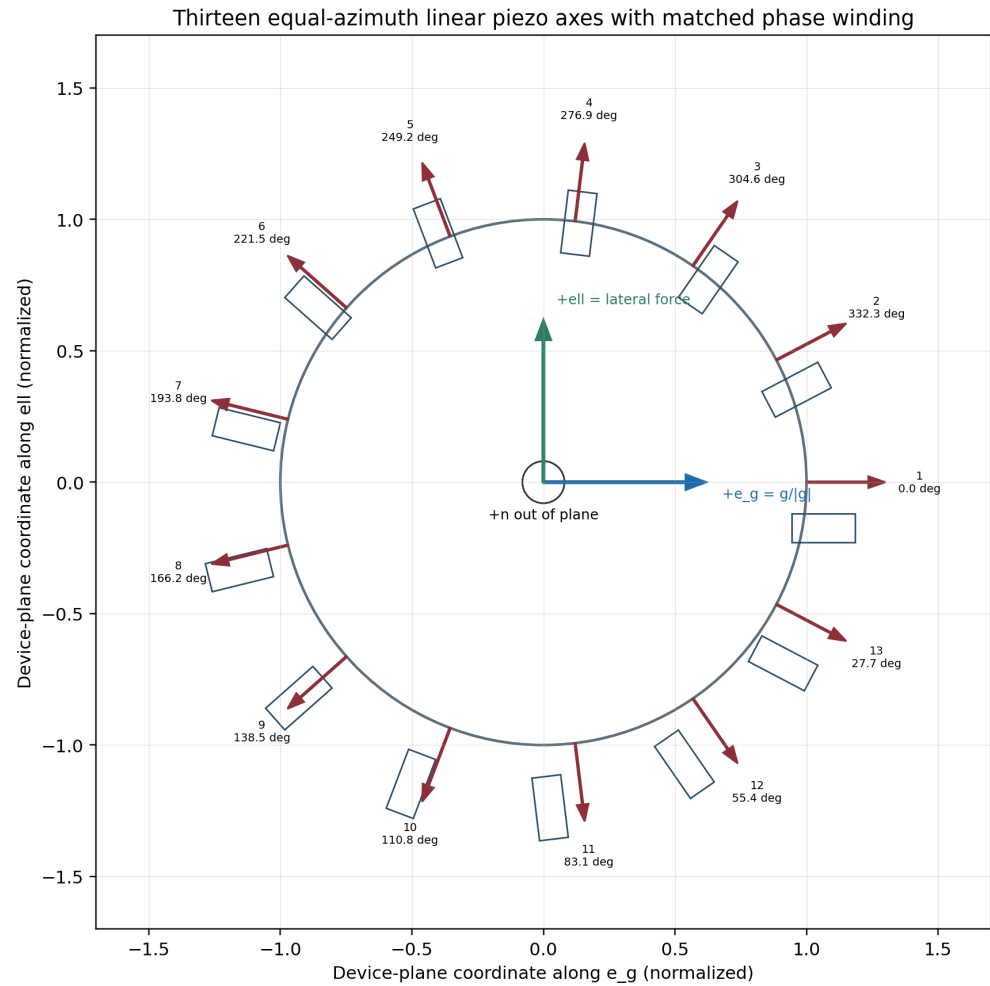


Figure 1. Optimized 13-element source-plane layout. Element azimuth and electrical phase are locked with opposite sign. The local device axes are e_g horizontally and ell vertically; n_{opt} is the oriented plane normal.

2. Source Geometry, Coordinate Lock, and Coupling Law

2.1 Cell3 coordinate map

The ordered Cell3 basis B maps integer address coordinates into the ICRS physical frame. The source uses nanometres for B and the centered half-step rule $p_L = B L / 2$ for L in Z^3 . The primitive translation associated with one full address increment is $B L$. For the finite centered registry used in this paper, each address coordinate takes the values -1, 0, and +1, producing 27 nodes.

The basis is skew rather than orthogonal. Consequently, equal address changes are not equal physical displacements, and a physical circle generally requires unequal Cell3 channel magnitudes and phases. The direct metric $M = B^T B$ measures physical squared lengths in address coordinates. The reciprocal metric $R = M^{-1}$ measures reciprocal-gradient squared lengths. The address coupling W and the physical response tensor G_{phase} obey the exact congruence identity shown below.

$$p_L = (1/2) B L \text{ qdot} = 2 B^{(-1)} \text{ v } G_{phase} = 4 B^{(-T)} W B^{(-1)}$$

Matrix	Numerical value
$B \text{ (nm)}$	0.327501242070 0.035330775844 -0.010129317627 -0.054897007886 0.207319850441 -0.389787702364 -0.037487502896 -0.547401085392 -0.689734408363
$M = B^T B \text{ (nm}^2\text{)}$	0.111676057905 0.020710333282 0.043937235098 0.020710333282 0.343877732397 0.296392758962 0.043937235098 0.296392758962 0.627770610069
$R = M^{(-1)} \text{ (nm}^{-2}\text{)}$	9.208025528684 0.001534802428 -0.645188035677 0.001534802428 4.903398312306 -2.315175584416 -0.645188035677 -2.315175584416 2.731171274746

Matrix	Numerical value
W (nm^2)	0.083353302633 0.137500576604 0.102559211222 0.137500576604 0.692400042849 0.550800904453 0.102559211222 0.550800904453 0.587747652353
G_phase	2.9764106378 1.2431507795 -2.0803600354 1.2431507795 2.5922286635 -2.3178839937 -2.0803600354 -2.3178839937 6.7735709133

2.2 Physical dimensions of the base cell

Quantity	Value
Edge a	0.334179679073151
Edge b	0.586410890414836
Edge c	0.792319765037693
Angle a-b (deg)	83.933483012312735
Angle a-c (deg)	80.448130018337906
Angle b-c (deg)	50.363231813756187
Cell volume (nm^3)	0.117914891916699
Face area ab (nm^2)	0.1948691654699
Face area ac (nm^2)	0.261106235799531
Face area bc (nm^2)	0.357809539128562

Cell3 centered 3 x 3 x 3 registry: all 27 nodes at true physical scale

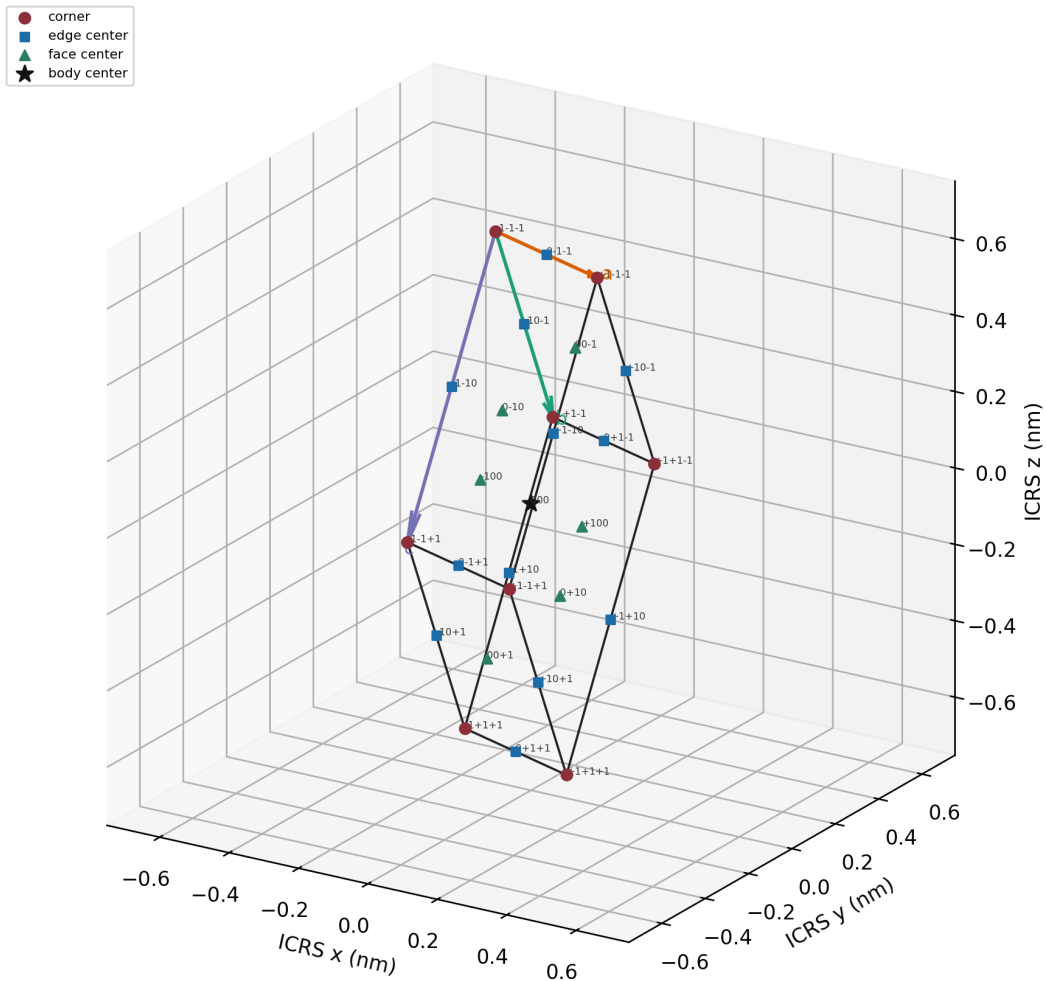


Figure 2. Dimensionally correct Cell3 centered registry. All 27 nodes use their supplied nanometre coordinates; the primitive cell edges connect the eight corner nodes. Marker shape distinguishes corners, edge centers, face centers, and the body center.

Table 1. Cell3 plane-family ledger

Channel	Face family	Unit normal (ICRS)	RA (deg)	Dec (deg)	Primitive spacing (nm)	Half-step spacing (nm)	Face area (nm^2)
a	bc	(-0.995965190, 0.083602163, -0.032619291)	175.201792089	-1.869279271	0.329546529709	0.164773264855	0.357809539129
b	ac	(0.089052594, 0.866576769, -0.491033948)	84.132672287	-29.408562676	0.451597379724	0.225798689862	0.261106235800
c	ab	(0.194092201, 0.913177169, 0.358379235)	78.000576203	21.000692601	0.605097741515	0.302548870757	0.194869165470

Three centered half-step plane families through the 27-node Cell3 registry

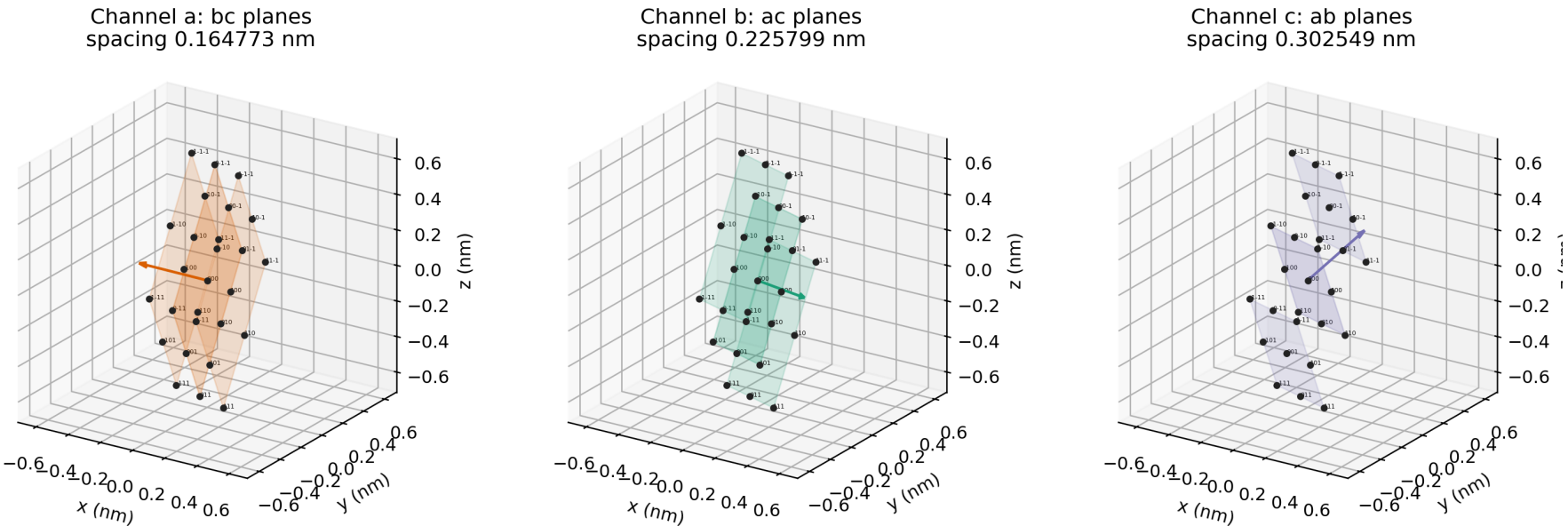


Figure 4. The three centered plane families rendered through all 27 nodes. Transparent parallelograms are the exact planes $L_i = -1, 0, +1$ under $p_L = B L/2$. All three subfigures use equal x, y, and z scales.

3. Earth-CMB Field State and the Lateral-Force Triad

The source Earth-CMB traversal vector V_E is expressed in ICRS Cartesian coordinates. The physical response tensor maps this velocity to $g = G_{\text{phase}} V_E$. The K-Field source law depends on g rather than on V_E alone. The angle between V_E and g is 31.640861271 degrees; anisotropy therefore rotates the response-gradient direction away from the kinematic translation direction.

For a circular or coherently circular oscillator with oriented normal n , the cycle velocity increment is $\Delta v = \pi \kappa_F \sigma n \times g$ and the mean acceleration is $\pi \kappa_F f \sigma n \times g$. The scalar orientation factor is $\sin(\beta)$, where β is the angle between n and g . The force vanishes for n parallel to g and reaches maximum magnitude for n perpendicular to g .

Object	ICRS vector	Magnitude	RA (deg)	Dec (deg)
Earth-CMB V_E (m/s)	(-359003.199065813038, 76679.886034001989, -44701.060100687995)	3.698125020004e+05	167.943299880	-6.942599922
Response $g = G_{\text{phase}} V_E$ (m/s)	(-880221.981635091361, -143911.436561599461, 266335.026948619285)	9.308252172704e+05	189.285391176	16.626277960
$e_g = g/ g $	(-0.945636157362, -0.154606293310, 0.286127859460)	1	189.285391176	16.626277960
n_{opt}	(-0.315865036407, 0.646162683024, -0.694768354089)	1	116.050867377	-44.008760899
ell	(0.077469585422, 0.747375863362, 0.659869519070)	1	84.082114314	41.289922334

$$g = G_{\text{phase}} V_E \quad \Delta v_{\text{cycle}} = \pi \kappa_F \sigma (n \times g) \quad \langle a \rangle = \pi \kappa_F f \sigma (n \times g)$$

Earth-CMB, response-gradient, plane-normal, and lateral-force triad

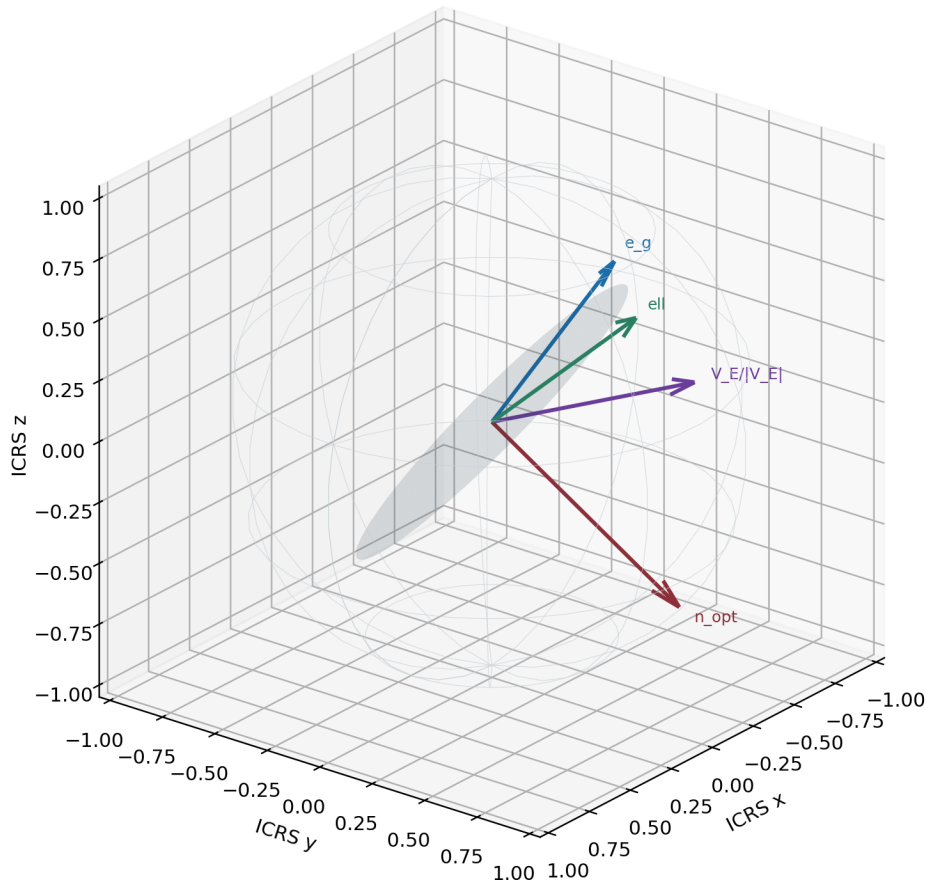


Figure 5. ICRS orientation triad. The oscillator plane is $\text{span}(e_g, ell)$; n_{opt} is perpendicular to that plane. The right-handed relations are $e_g \times ell = n_{\text{opt}}$ and $n_{\text{opt}} \times e_g = ell$.

3.1 Optimization of the plane normal

Maximum force magnitude alone leaves a great circle of admissible normals satisfying $n \cdot g = 0$. The thruster objective additionally requires the force to be lateral to V_E . Since $n \times g$ must be perpendicular to both n and g , the purely lateral direction is parallel to $V_E \times g$. The corresponding normal is the normalized projection of V_E into the plane perpendicular to g .

$$\mathbf{n}_{\text{opt}} = [\mathbf{V}_E - (\mathbf{V}_E \cdot \mathbf{e}_g)\mathbf{e}_g] / |\mathbf{V}_E - (\mathbf{V}_E \cdot \mathbf{e}_g)\mathbf{e}_g| \quad \mathbf{ell} = (\mathbf{V}_E \times \mathbf{g}) / |\mathbf{V}_E \times \mathbf{g}| = \mathbf{n}_{\text{opt}} \times \mathbf{e}_g$$

The resulting plane normal is $\mathbf{n}_{\text{opt}} = (-0.315865036407, 0.646162683024, -0.694768354089)$. Its ICRS right ascension is 116.050867377 degrees and declination is -44.008760899 degrees. The antipode describes the same unoriented physical plane. The oriented normal is fixed by the phase winding and desired force sign.

The angle between $\mathbf{n}_{\text{opt}}$ and $\mathbf{g}$ is 90.000000000000 degrees. The angle between $\mathbf{n}_{\text{opt}}$ and $\mathbf{V}_E$ is 58.359138728896 degrees, so $\mathbf{V}_E$ lies 31.640861271104 degrees above the device plane. The positive force direction $\mathbf{ell}$ has right ascension 84.082114314 degrees and declination 41.289922334 degrees and is perpendicular to $\mathbf{V}_E$ to numerical precision.

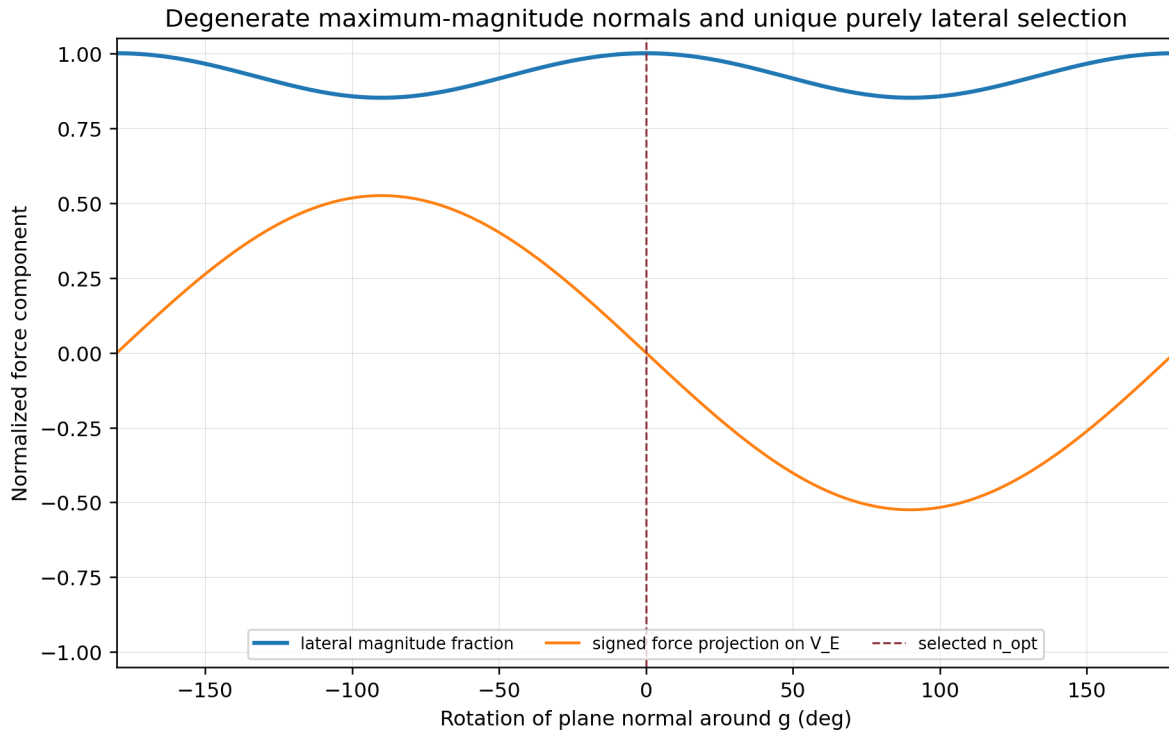


Figure 6. All normals on the great circle perpendicular to $\mathbf{g}$ produce the same total force magnitude. The selected $\mathbf{n}_{\text{opt}}$ is the member whose force has zero projection on $\mathbf{V}_E$ and therefore lies entirely in the lateral subspace.

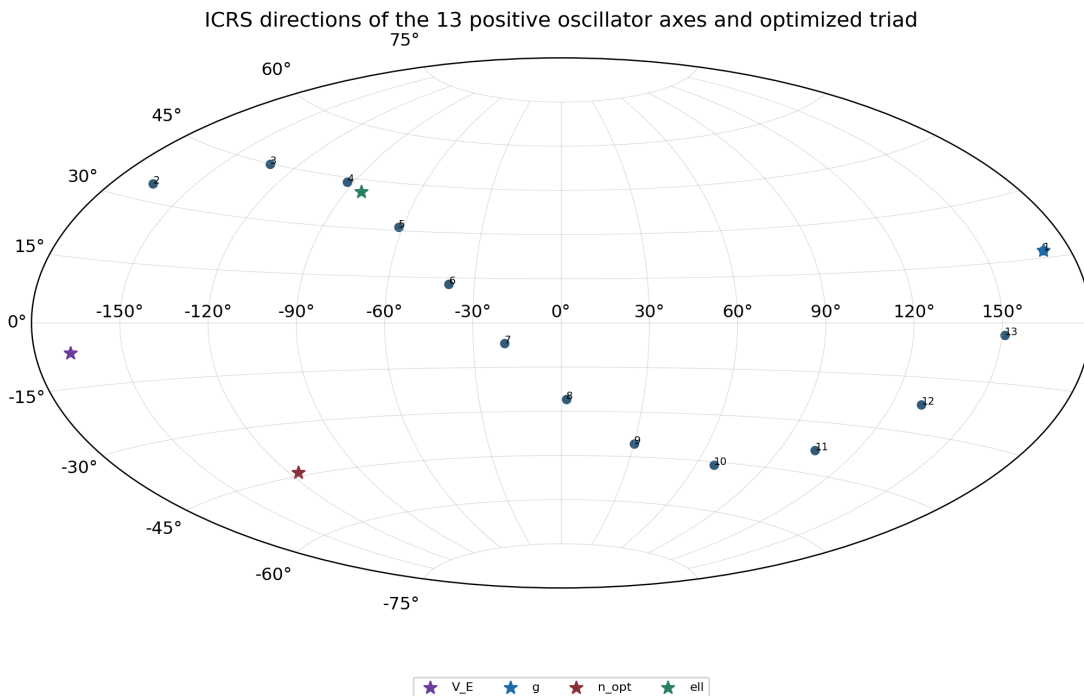


Figure 7. ICRS sky map of the thirteen positive oscillator-axis directions and the optimized triad. Right ascension increases to the left in this astronomical Aitoff rendering.

4. Single Linear Element in the Fixed Earth-CMB Frame

A local linear piezo element closes in the apparatus frame but does not close in the fixed Earth-CMB frame. For axis d_j , amplitude A , phase ϕ_j , and mount position r_{j0} , its absolute trajectory is $r_j(t) = r_{j0} + V_E t + A \cos(\omega t + \phi_j)d_j$. The local displacement returns after one period, while the absolute endpoint advances by V_E/f .

At the 100 MHz design point, the period is 10.000000 ns and Earth advances 3.698125020 mm during one cycle. The corresponding Cell3 half-step address displacement is $(-2.217734842862e7, 2.499065863112e6, 5.181751775713e5)$. The apparatus is therefore locally periodic and address-open at every carrier cycle.

For a forward half-stroke from $-A$ to $+A$ and a reverse half-stroke from $+A$ to $-A$, define $D = |V_E|/(2f)$ and $\mu_j = d_j \cdot V_E/|V_E|$. The two CMB-frame path lengths are not equal unless $\mu_j = 0$. Their exact difference is retained in the element ledger and is plotted below for a reference amplitude of one micrometre.

$\Delta r_{F,j} = (V_E/2f) + 2A d_j$	$\Delta r_{R,j} = (V_E/2f) - 2A d_j$
$L_{F,j} = \sqrt{D^2 + 4A^2 + 4AD \mu_j}$	$L_{R,j} = \sqrt{D^2 + 4A^2 - 4AD \mu_j}$
For $A \ll D$: $L_{F,j} - L_{R,j} \approx 4A \mu_j$	

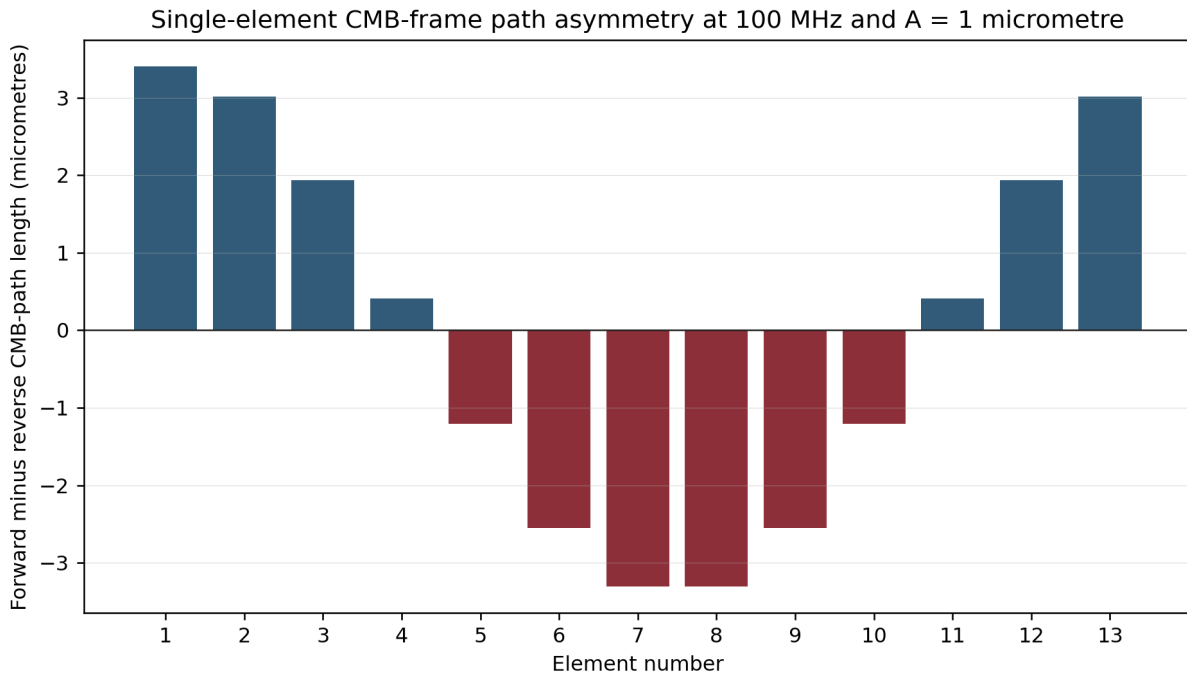


Figure 8. Forward-minus-reverse CMB-frame path length for the thirteen optimized axes at 100 MHz and $A = 1$ micrometre. Negative values mean that the branch named reverse is longer; exchanging the positive axis reverses the labels but not the physical pair.

4.1 Why phase ordering is the residual-generating extension

One isolated reciprocal coordinate retraces one line in local displacement space. Under a constant address-independent differential kernel, its signed local loop integral cancels over a cycle. The thirteen-element construction introduces nonparallel coordinates and a fixed phase ordering. The composite coherent state no longer lies on one line; it closes as an oriented circle. Its pairwise cross terms are proportional to $\sin(\phi_j - \phi_i)(d_i \times d_j)$, so they vanish for equal or opposite phase and survive for a circulating phase sequence.

The K-Field continuation used here couples to that coherent composite mode. The degree to which distributed linear elements share one physical K-Field address state is included in H_{13} . In the ideal coherent limit $H_{13} = 1$, the array is normalized to the same circularity convention as the rotating-tube source. If the elements behave as entirely independent one-dimensional sources with no cross-element coherence, H_{13} for this mechanism is zero. The experiment therefore measures H_{13} rather than assuming it silently.

5. Exact Coherent Synthesis from Thirteen Linear Axes

Let the optimized device-plane basis be $\mathbf{e}_g$ and $\mathbf{e}_\ell$, with oriented normal $\mathbf{n}_{\text{opt}} = \mathbf{e}_g \times \mathbf{e}_\ell$. Place element j at azimuth $\theta_j = 2\pi(j-1)/13$ and point its positive linear axis along $\mathbf{d}_j = \cos(\theta_j)\mathbf{e}_g + \sin(\theta_j)\mathbf{e}_\ell$. The scalar displacement is $x_j(t) = A \cos(\omega t + \phi_j)$.

Choose $\phi_j = \phi_0 - \theta_j$. Expanding $\cos(\omega t + \phi_0 - \theta_j)$ and using the exact sums of the thirteenth roots of unity gives $\sum \cos(\theta_j) = \sum \sin(\theta_j) = \sum \sin(\theta_j)\cos(\theta_j) = 0$ and $\sum \cos^2(\theta_j) = \sum \sin^2(\theta_j) = 13/2$. The coherent vector therefore becomes an exact circle.

$$\begin{aligned} \theta_j &= 2\pi(j-1)/13 \quad \mathbf{d}_j = \cos(\theta_j)\mathbf{e}_g + \sin(\theta_j)\mathbf{e}_\ell \quad \phi_j = \phi_0 - \theta_j \\ \mathbf{u}_{13}(t) &= (2/13) \sum_j x_j(t)\mathbf{d}_j = A[\cos(\omega t + \phi_0)\mathbf{e}_g + \sin(\omega t + \phi_0)\mathbf{e}_\ell] \end{aligned}$$

The two quadrature vectors are exactly orthogonal and equal in magnitude. Their cross product points along $\mathbf{n}_{\text{opt}}$, and the normalized mechanical circularity is exactly +1 for the negative phase winding. Reversing the winding to $\phi_j = \phi_0 + \theta_j$ changes the sign of the second quadrature, reverses the loop normal, and sends the force from $+\mathbf{e}_\ell$ to $-\mathbf{e}_\ell$.

A common phase offset ϕ_0 changes only the time origin. Adding 180 degrees to every drive also changes only the time origin and does not reverse chirality. Reversing one physical axis while keeping its positive direction convention requires adding 180 degrees to that element phase so that the physical displacement remains unchanged.

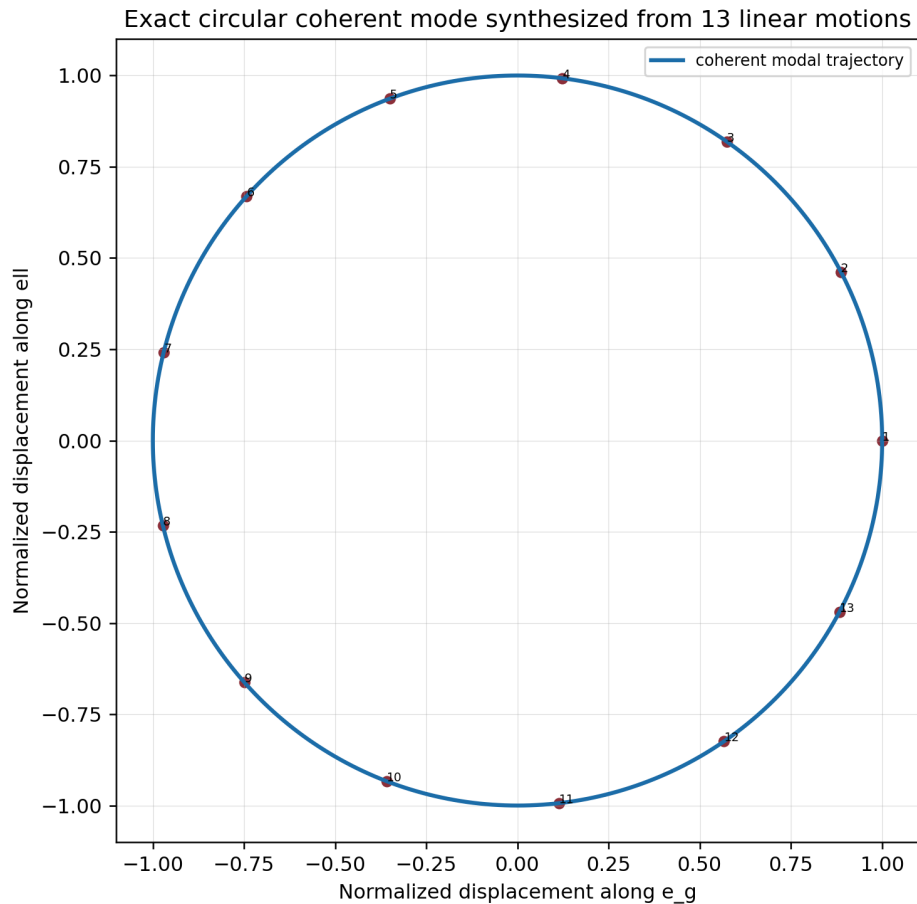


Figure 9. Direct numerical evaluation of the thirteen-element sum. The normalized coherent coordinate traces a unit circle in $\text{span}(\mathbf{e}_g, \mathbf{e}_\ell)$, and its normal is $\mathbf{n}_{\text{opt}}$.

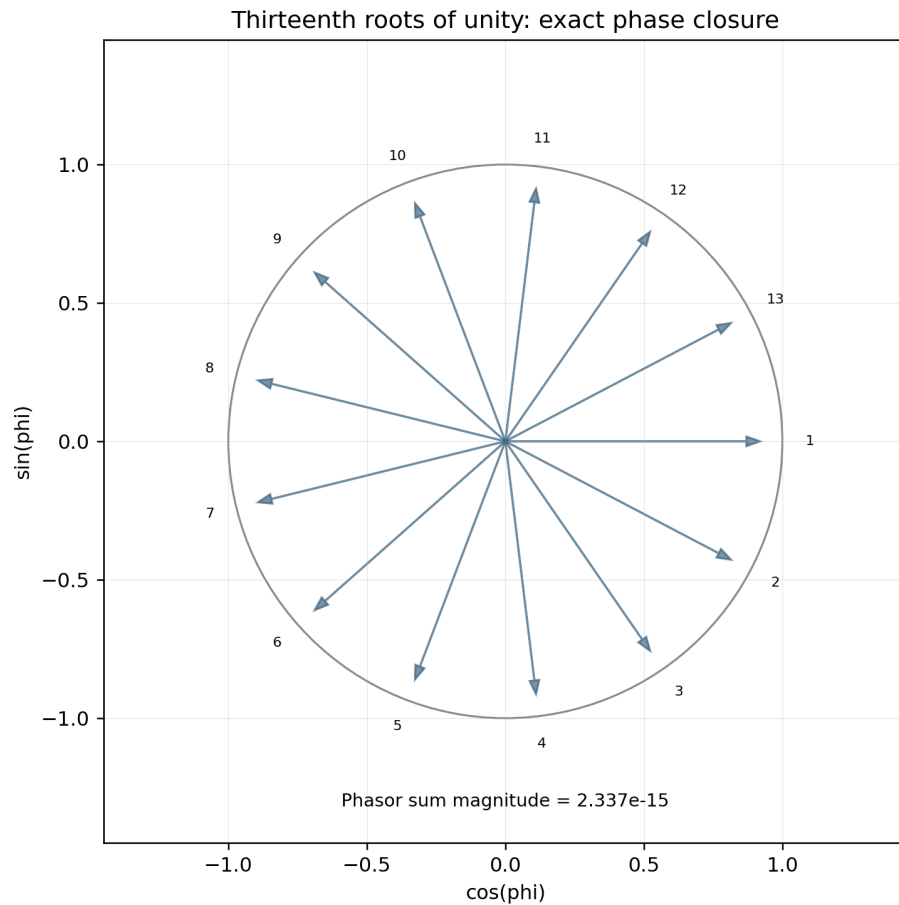


Figure 10. Electrical phase phasors. The vector sum of the thirteen phase roots is zero to floating-point precision. This exact closure removes the common normal component of a conical installation and fixes the traveling-wave phase order.

5.1 Proof of optimal circular amplitude

Write the complex coherent coefficient as $C = \sum_j \exp(i \phi_j) d_j$. The real trajectory is $\text{Re}[C \exp(i \omega t)]$. A circle requires the real and imaginary quadratures of C to be orthogonal and equal. Since each term has unit norm, the triangle inequality gives $|C| \leq 13$. For a circular trajectory of radius R_c , $|C| = \sqrt{2} R_c$, so $R_c \leq 13/\sqrt{2}/\sqrt{2} = 13/2$ in the unnormalized sum.

The uniform azimuth and matched phase construction attains $u_{\cos} = (13/2)e_g$ and $u_{\sin} = (13/2)e_{\ell}$. It therefore reaches the bound exactly. No other arrangement of thirteen equal-stroke unit axes can produce a larger equal-quadrature coherent circle. Nonuniform designs may redistribute actuator load or accommodate packaging, but they cannot exceed the ideal circular amplitude under the same stroke constraint.

6. General Off-Normal Installation and Tilt Optimization

The user-specified architecture permits each linear axis to be offset from the plane normal. Let α be the common angle from n_{opt} . A conical axis is $d_j(\alpha) = \cos(\alpha)n_{\text{opt}} + \sin(\alpha)[\cos(\theta_j)e_g + \sin(\theta_j)e_{\ell}]$. The same thirteenth-root phase law cancels the coherent normal term exactly because $\sum_j \cos(\omega t - \theta_j) = 0$.

The surviving circular coordinate has radius $A \sin(\alpha)$. Its oriented area therefore scales as $\sin^2(\alpha)$ for fixed individual stroke amplitude. The unconstrained optimum is $\alpha = 90$ degrees. Every out-of-plane component consumes actuator stroke but contributes no coherent circular area in the selected plane.

$$d_j(\alpha) = \cos(\alpha)n_{\text{opt}} + \sin(\alpha)[\cos(\theta_j)e_g + \sin(\theta_j)e_{\ell}]$$

$$u_{13}(t, \alpha) = A \sin(\alpha)[\cos(\omega t)e_g + \sin(\omega t)e_{\ell}] \quad \eta_{\alpha} = \sin^2(\alpha)$$

alpha from normal	Relative residual $\sin^2(\alpha)$
0 deg	0.000000000
30 deg	0.250000000

alpha from normal	Relative residual $\sin^2(\alpha)$
45 deg	0.500000000
60 deg	0.750000000
75 deg	0.933012702
80 deg	0.969846310
85 deg	0.992403877
90 deg	1.000000000

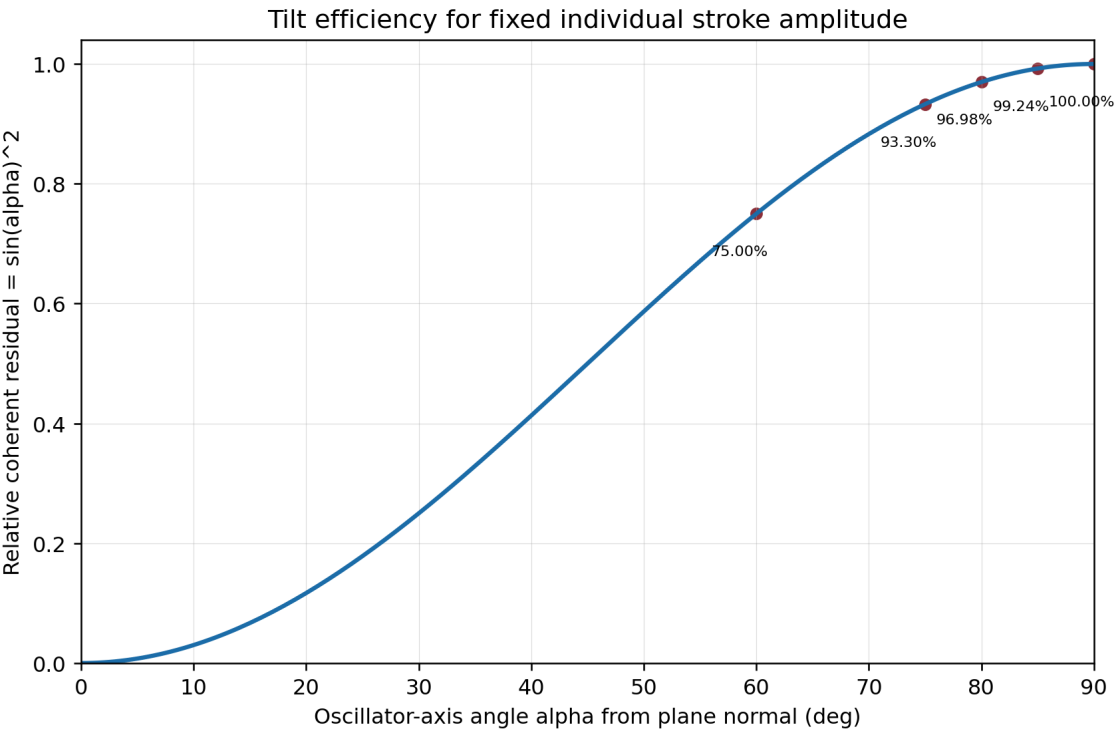


Figure 11. Residual efficiency versus common axis tilt. The curve is an exact geometric result for fixed individual stroke amplitude and the coherent-mode normalization used in this report.

7. Exact Thirteen-Element Orientation and Timing Ledger

The following tables are the complete installation and drive specification for the positive-axis convention used in the figures. Device azimuth is measured in the optimized plane from +e_g toward +ell. The modulo phase is the direct synthesizer setting for $x_j = A \cos(\omega t + \phi_j)$. The delay is measured from element 1 at 100 MHz and is equivalent to the negative phase winding.

j	theta (deg)	phase mod 360 (deg)	signed phase (deg)	delay (ns)	d dot e _g	d dot V _{hat}	F/R path difference at A=1 um (um)
1	0.00000000	0.00000000	-0.00000000	0.00000000	1.00000000	0.851353030	3.405411571
2	27.692307692	332.307692308	-27.692307692	0.769230769	0.885456026	0.753835670	3.015341919
3	55.384615385	304.615384615	-55.384615385	1.538461538	0.568064747	0.483623643	1.934493706
4	83.076923077	276.923076923	-83.076923077	2.307692308	0.120536680	0.102619268	0.410476834
5	110.769230769	249.230769231	-110.769230769	3.076923077	-0.354604887	-0.301893945	-1.207575138
6	138.461538462	221.538461538	-138.461538462	3.846153846	-0.748510748	-0.637246893	-2.548986688
7	166.153846154	193.846153846	-166.153846154	4.615384615	-0.970941817	-0.826614258	-3.306456420
8	193.846153846	166.153846154	-193.846153846	5.384615385	-0.970941817	-0.826614258	-3.306456420
9	221.538461538	138.461538462	-221.538461538	6.153846154	-0.748510748	-0.637246893	-2.548986688
10	249.230769231	110.769230769	-249.230769231	6.923076923	-0.354604887	-0.301893945	-1.207575138
11	276.923076923	83.076923077	-276.923076923	7.692307692	0.120536680	0.102619268	0.410476834
12	304.615384615	55.384615385	-304.615384615	8.461538462	0.568064747	0.483623643	1.934493706
13	332.307692308	27.692307692	-332.307692308	9.230769231	0.885456026	0.753835670	3.015341919

j	d _x	d _y	d _z	RA (deg)	Dec (deg)	Angle to V _E (deg)	Angle to g (deg)
1	-0.945636157362	-0.154606293310	0.286127859460	189.285391176	16.626277960	31.640861271	0.000000000
2	-0.801317322138	0.210425807916	0.560010293303	165.286344092	34.056509596	41.076263052	27.692307692
3	-0.473426345332	0.527251892453	0.705601117806	131.921080659	44.878127926	61.077663443	55.384615385
4	-0.037079098217	0.723290922504	0.689547229636	92.934666575	43.594278850	84.109980352	83.076923077
5	0.407762523448	0.753632718810	0.515526381100	61.583834725	31.032645524	107.571393419	110.769230769
6	0.759190665062	0.611326341495	0.223404651421	38.842208754	12.909083648	129.586832266	138.461538462
7	0.936697374550	0.328972466625	-0.119896391581	19.351511692	-6.886123083	145.752495187	166.153846154
8	0.899618003955	-0.028745035801	-0.435730616179	358.169879803	-25.831793234	145.752495187	166.153846154
9	0.656446990227	-0.379877396941	-0.651744207735	329.942587713	-40.673237110	129.586832266	138.461538462
10	0.262891882081	-0.643984424460	-0.718451055667	292.206612701	-45.926744357	107.571393419	110.769230769
11	-0.190888588058	-0.760562381189	-0.620569425020	255.910734589	-38.357728839	84.109980352	83.076923077
12	-0.600938783131	-0.702904662158	-0.380522817773	229.471662581	-22.366070949	61.077663443	55.384615385
13	-0.873321145086	-0.484219955946	-0.053303018772	209.006530882	-3.055486058	41.076263052	27.692307692

8. Frequency Optimization and the 100 MHz Design Point

At maximum orientation and unit circularity, the source law contains frequency as a linear multiplier. A practical resonator introduces two additional frequency-dependent quantities: $\eta_m(f)$, the fraction of installed mass participating in the commanded mode, and $H_{13}(f)$, the coherent K-Field transfer of the distributed linear array. The correct optimization is therefore not to maximize frequency alone, but to maximize $f \eta_m(f) H_{13}(f)$.

No finite global frequency optimum follows from geometry without measured or otherwise specified transfer functions. In the ideal continuation $\eta_m = H_{13} = 1$, the objective is monotonic. The previous design response selected a 50-100 MHz high-overtone piezoelectric implementation window. Under that explicit constraint, the nominal optimum is the upper endpoint, 100 MHz. This is an engineering design point, not an assertion that the underlying response peaks at 100 MHz.

$$f_{\text{opt}} = \arg \max_f [f \eta_m(f) H_{13}(f)] \text{ constrained ideal result over 50-100 MHz: } f_{\text{opt}} = 100 \text{ MHz}$$

Quantity	100 MHz value
Frequency	100000000 Hz
Period	10.000000000 ns
Adjacent phase increment	-27.692307692308 deg
Adjacent delay	0.769230769231 ns
Earth-CMB translation per cycle	3.698125020004 mm
Earth-CMB translation per half-cycle	1.849062510002 mm
Cell3 address shift per cycle	(-2.217735e+07, 2.499066e+06, 5.181752e+05)

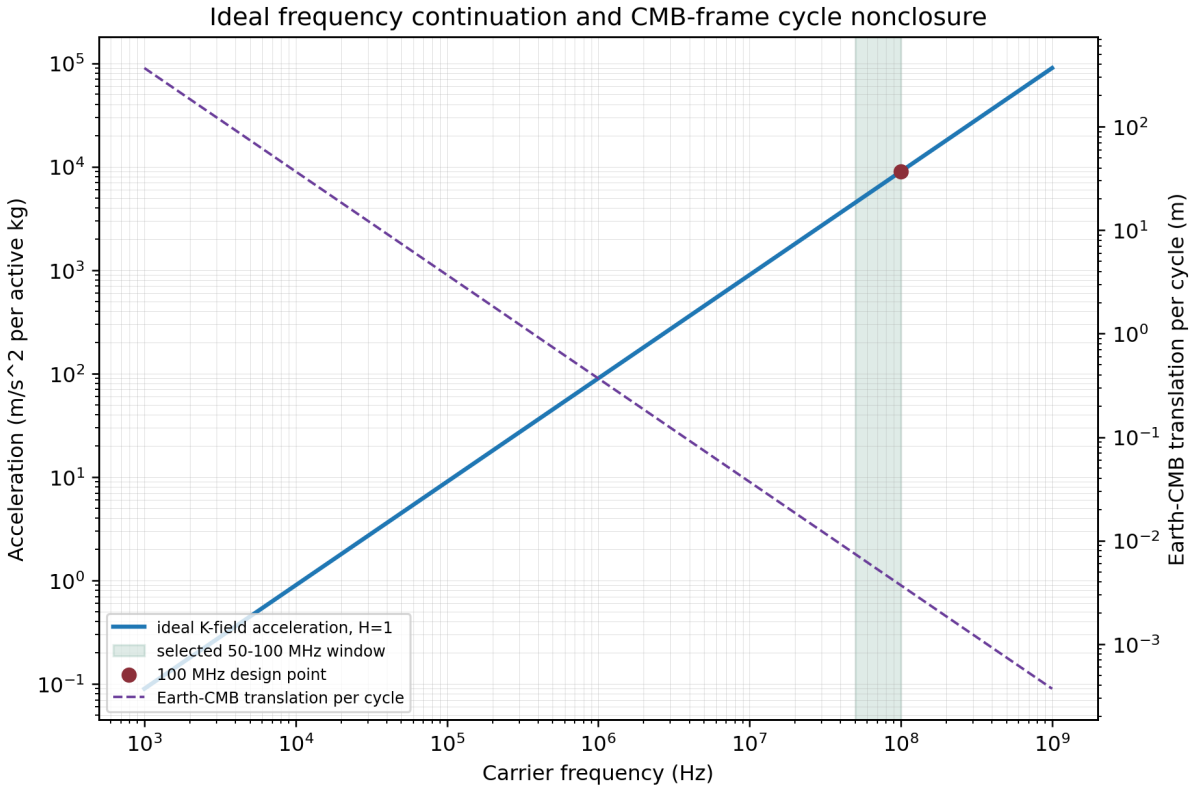


Figure 12. Ideal $H = 1$ acceleration rises linearly with frequency, while the Earth-CMB translation per cycle falls as $1/f$. The selected 50-100 MHz design window is highlighted. The actual optimum is the maximum of $f \eta_m H_{13}$.

8.1 Timing implementation

At 100 MHz, the thirteen channels form a uniformly sampled traveling phase pattern. The exact adjacent delay is $T/13 = 0.769230769$ ns. A direct digital or RF phase network may implement the phases as absolute phase offsets; a physical delay line may implement the same relation in time. Closed-loop control must regulate the measured mechanical phase rather than assuming that equal electrical phase produces equal strain phase.

Cable length, electrode impedance, piezoelectric dispersion, and cross-axis coupling create channel-dependent phase shifts. Each channel therefore requires complex calibration. A laser-vibrometry or strain-sensor matrix can estimate the thirteen scalar modal coordinates and apply a digital inverse transfer matrix so that the mechanical phase ledger, not merely the voltage ledger, remains locked.

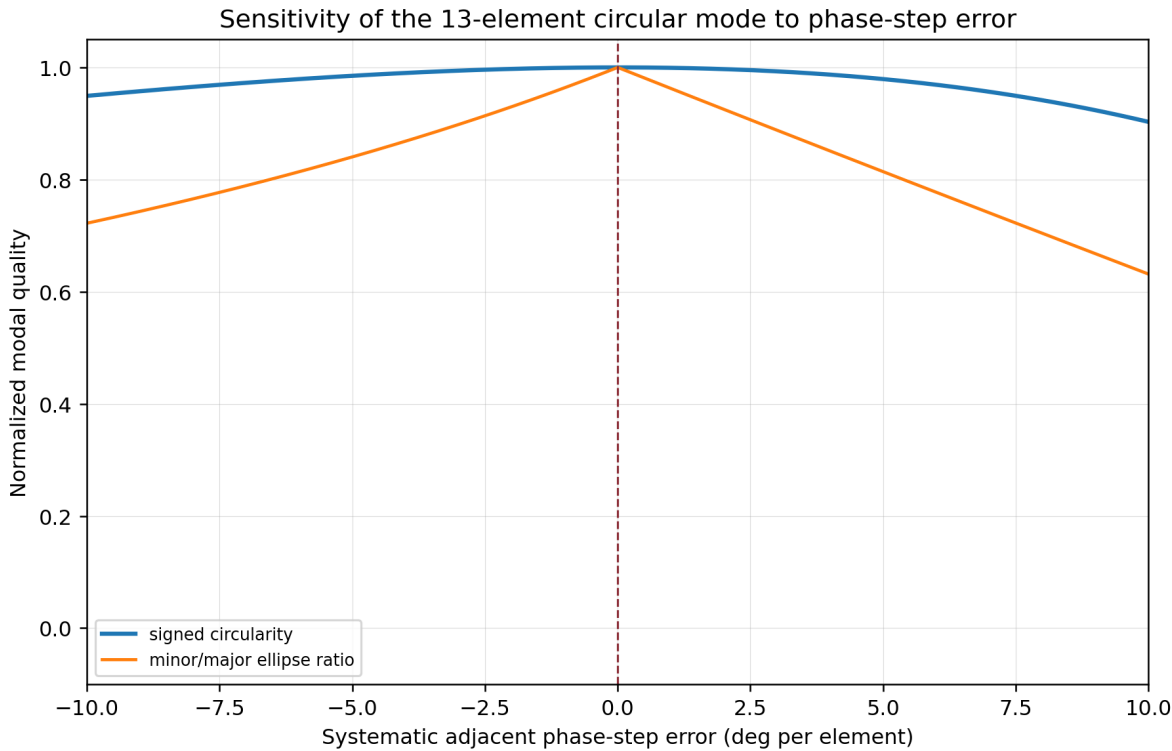


Figure 13. Deterministic sensitivity to a systematic error in the adjacent phase step. The ideal point has unit circularity and equal ellipse axes. Phase error changes both chirality purity and modal ellipticity.

9. Force Law, Scaling, and Dimensional Closure

The distributed engineering force law is the mass integral of the circular source response. For the coherent thirteen-element source plane it is convenient to collect unresolved spatial coherence, localization, material, and boundary effects into H_{13} . The normalized circularity C_{13} equals +1 for the optimized negative winding and -1 for the reversed winding. The orientation factor equals unity for n_{opt} .

$$F_K = M_{active} \pi \kappa_F f C_{13} H_{13} (n_{opt} \times g)$$

$$|F_K| = 8.989366088879e-5 M_{active} f |C_{13}| H_{13} \sin(\beta)$$

Within the source convention, G_{phase} and κ_F are dimensionless. The vector g has units of velocity. Multiplication by f gives acceleration, and multiplication by active mass gives force. The maximum coefficient is $8.989366088879e-5 \text{ N}/(\text{kg Hz})$. One newton at $H_{13} = 1$ therefore requires $M_{active} f = 11124.254926 \text{ kg Hz}$.

At 100 MHz, the ideal acceleration is $8989.366088879 H_{13} \text{ m/s}^2$. This value is independent of active mass; mass controls absolute force. The table below evaluates the equation for selected active masses and does not represent measured output.

Active mass	Ideal force at 100 MHz, $H_{13}=1$	Equivalent note
1 mg	0.008989366089 N	$M f = 100.000000 \text{ kg Hz}$
13 x 1 mg	0.116861759155 N	$M f = 1300.000000 \text{ kg Hz}$
100 mg	0.898936608888 N	$M f = 10000.000000 \text{ kg Hz}$
1 g	8.989366088879 N	$M f = 100000.000000 \text{ kg Hz}$
10 g	89.893660888791 N	$M f = 1000000.000000 \text{ kg Hz}$

9.1 Source calibration against the rotating tube

Source quantity	Value
Tube mass	0.022619467105847 kg
Spin frequency	833.333333333333 Hz
Mean acceleration magnitude	0.074911384073992 m/s^2
Force magnitude	0.001694455587915 N
Cycle velocity increment	8.989366088879059e-05 m/s
Acceleration per hertz	8.989366088879062e-05 m/s^2/Hz

10. Cell3 Channel Synthesis of the Optimized Physical Circle

The optimized device plane is physically orthonormal, but Cell3 address coordinates are skew. The exact complex command that produces a physical circle in span(e_g, ell) is $Q = 2 B^{(-1)}(e_g - i \ell)$. Its three channels therefore require unequal amplitudes and phases. This channel-space result is independent of the thirteen-element packaging; it is the address representation of the same physical circular order parameter.

$$Q = 2 B^{(-1)} (e_1 - i e_2)$$

Channel	Q real per nm	Q imag per nm	Q per nm	Phase (deg)	Relative magnitude	Relative phase (deg)
a	-5.580771783559	-0.219691268566	5.585094265422	-177.745672651	1.000000000000	0.000000000
b	-1.588530331304	-1.463866302932	2.160169707762	-137.338737757	0.386774082066	40.406934894
c	0.734364852431	3.087125555117	3.173268965837	76.619183540	0.568167485638	-105.635143809

Exact complex Cell3 command for the optimized physical circular plane

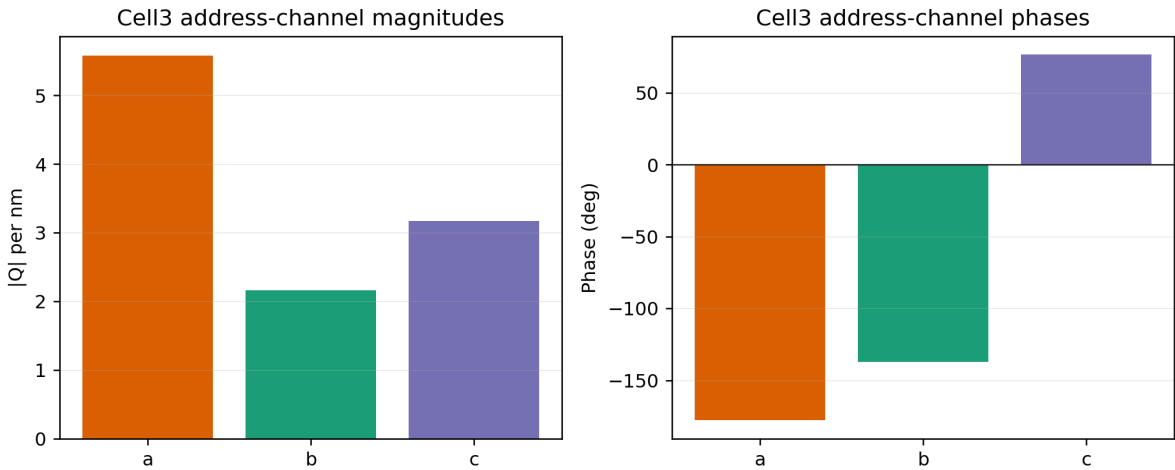


Figure 14. Exact Cell3 complex-channel magnitudes and phases for the optimized physical circular plane. Equal physical quadratures do not correspond to equal address-channel commands.

11. Piezoelectric Mechanical Architecture

The source plane is most naturally implemented as an annular support carrying thirteen identical linear piezoelectric resonator cells. Each cell has one measured scalar displacement axis. The positive axis points radially outward in the local e_g-ell drawing, and the placement azimuth equals the axis azimuth. The support plane is installed with normal n_opt in ICRS. The device may be rotated 180 degrees about any in-plane axis if the phase winding and positive-axis conventions are transformed consistently.

At 100 MHz, a high-overtone bulk-acoustic, thickness-shear, guided-wave, or MEMS resonator is more appropriate than a macroscopic flexural stack. The exact material, electrode pattern, overtone number, and substrate thickness are not fixed by the K-Field source data. They must be selected to maximize active mass, mechanical quality factor, strain uniformity, phase controllability, and the measured product $f \eta_m H_{13}$.

The thirteen cells should be fabricated and mounted as symmetrically as possible. Equal resonance frequencies, damping, effective masses, and electrical transfer functions reduce the calibration burden. The electrical drive should be multi-channel and phase coherent. Each element requires amplitude and phase feedback because piezoelectric transfer is complex and can drift with temperature, load, and electrode impedance.

A stationary force-balance experiment should modulate chirality at a low reference frequency by alternating the phase winding between $\phi_j = \phi_0 - \theta_j$ and $\phi_j = \phi_0 + \theta_j$. This reverses the K-Field residual while preserving carrier frequency, nominal RMS voltage, and mean heating. Synchronous detection at the chirality-switch frequency rejects much of the 100 MHz carrier background.

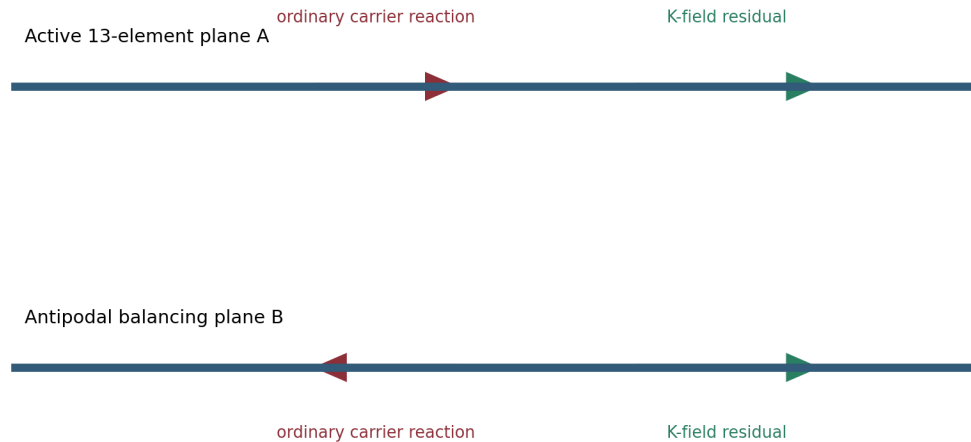
11.1 Ordinary reaction force and required balancing

The ideal thirteen-element source plane intentionally creates a nonzero coherent vector coordinate. It therefore also creates a conventional carrier-frequency support reaction. The roots-of-unity phase law cancels the common normal component and creates circular isotropy; it does not by itself cancel the in-plane first harmonic. For equal element masses totaling M_{active} and equal scalar stroke A , the center-of-mass circle has radius $A/2$, so the conventional carrier-force amplitude is $M_{\text{active}} A \omega^2/2$.

Mechanical balancing is therefore mandatory. The cleanest extension is a second antipodal thirteen-element plane driven so that every physical displacement and velocity is the negative of the first plane. Ordinary reactions then cancel. The oriented-loop parity is unchanged because both displacement and velocity reverse, so the two K-Field contributions add under the source symmetry. The active element count of the source plane remains thirteen; the full reaction-balanced package contains two source planes.

$$|F_{\text{carrier}}| = (1/2) M_{\text{active}} A \omega^2 \quad F_K/F_{\text{carrier}} = \kappa_F |g| / (A \omega) = 4.554066061598e-6 / [A(m) f(\text{Hz})]$$

Reaction-balanced antipodal extension of the 13-element source plane



Displacement and velocity reverse together in plane B; ordinary reactions cancel while oriented-loop parity is retained.

Figure 15. Antipodal balancing principle. The mirror plane reverses ordinary displacement and velocity together. Conventional carrier reactions cancel, while the oriented-loop parity and predicted K-Field residual retain the same sign.

12. Measurement, Controls, and Parameter Identification

The experiment should measure mechanical state and force state independently. The mechanical dataset must include the complex displacement of every element, carrier-frequency support acceleration, harmonics, temperature, voltage, current, and acoustic emission. The force dataset should be acquired on a calibrated balance or flexure whose sensitive axis is aligned with ℓ and whose orthogonal channels are recorded simultaneously.

The primary observable is the force component that changes sign under phase-winding reversal. A complete orientation test rotates the device normal from n_{opt} toward g . The predicted magnitude follows $\sin(\beta)$ and reaches a null when n is parallel to g . A second null sets all phases equal or opposite so that the coherent trajectory becomes nonchiral. A third control preserves the 100 MHz electrical

load while randomizing the spatial phase order.

The transfer factor H_{13} must be inferred rather than assumed. At each operating point, compute $H_{13} = F_{\text{measured}}/[M_{\text{active}} \pi \kappa_F f C_{13} |n \times g|]$. Mapping H_{13} against amplitude, frequency, overtone number, material, temperature, pressure, mounting condition, and phase-code family identifies the coherence and localization structure of the device.

The strongest internal consistency test is vectorial. The force should follow $n \times g$, not merely a fixed laboratory axis. Phase-winding reversal should reverse the vector. Reversing the device plane normal while holding the physical chirality convention fixed should transform the sign according to the same cross product. Daily and seasonal changes in the laboratory mapping of the fixed ICRS directions can be calculated and used as an additional orientation signature when geographic location and time are specified.

Control	Configuration	Predicted K-Field result
Chirality reversal	$\phi_j = \phi_0 - \theta_j$ versus $\phi_0 + \theta_j$	Equal magnitude, opposite sign
Linear/null phase	All ϕ_j equal or arranged for zero coherent area	Zero coherent circular residual
Orientation null	n parallel to g	Zero $n \times g$ response
Antipodal plane	Second plane with all physical displacements reversed	Carrier reaction cancels; K residual adds
Frequency sweep	Hold measured C_{13} and M_{active} ; sweep overtone	Tests $f \eta_m H_{13}$
Amplitude sweep	Hold frequency and circularity; reduce stroke	Locates small-amplitude transition in H_{13}
Phase-code perturbation	Systematic and randomized phase errors	Tests coherent cross-element mechanism
Dummy electrical load	Matched impedance without piezo motion	Rejects electromagnetic and thermal pickup

13. Tolerances and Calibration Requirements

Plane-normal error and phase error affect different parts of the response. A small angular error ϵ that tilts n away from perpendicular to g reduces the total magnitude by $\cos(\epsilon)$. A five-degree error retains 99.619 percent of the ideal magnitude, while a ten-degree error retains 98.481 percent. Rotation of n around g leaves total magnitude unchanged but introduces a component along V_E ; the selected n_{opt} is required when the design objective is a purely lateral force.

Element tilt error reduces projected coherent area. For a common tilt α , the exact factor is $\sin^2(\alpha)$. Individual tilt errors also introduce unequal quadratures and residual normal motion. Their effect should be computed from the measured axis vectors rather than from nominal drawings.

Phase accuracy is more stringent at high frequency because delay errors scale directly into phase. At 100 MHz, one picosecond corresponds to 0.036 degrees, ten picoseconds to 0.36 degrees, and one hundred picoseconds to 3.6 degrees. The phase-step sensitivity figure shows that systematic error accumulates around the thirteen-element ring; closed-loop mechanical calibration is therefore preferable to open-loop cable trimming.

Amplitude mismatch changes the relative contribution of the axes and converts the circular order parameter into an ellipse. The appropriate real-time circularity estimator is $C_{13} = 2(u_{\cos} \times u_{\sin}) \cdot n_{\text{opt}} / (|u_{\cos}|^2 + |u_{\sin}|^2)$, where $u_{\cos}$ and $u_{\sin}$ are reconstructed from the measured element phasors. This measured C_{13} should be used in every force normalization.

$$C_{13} = 2[(u_{\cos} \times u_{\sin}) \cdot n_{\text{opt}}] / (|u_{\cos}|^2 + |u_{\sin}|^2)$$

14. Consolidated Findings and Final Design Specification

The optimized thirteen-element array is a coherent circular-mode synthesizer constructed from one-dimensional piezo coordinates. Its mathematical optimum uses equal azimuth steps of 360/13 degrees, a matched negative phase winding, and axes lying in the optimized device plane. The construction reaches the maximum possible equal-quadrature coherent amplitude for thirteen equal unit-stroke linear actuators.

The optimized plane normal is $n_{\text{opt}} = (-0.315865036407, 0.646162683024, -0.694768354089)$, with ICRS RA 116.050867377 degrees and Dec -44.008760899 degrees. The positive force direction is $ell = (0.077469585422, 0.747375863362, 0.659869519070)$, with RA 84.082114314 degrees and Dec 41.289922334 degrees. The response-gradient axis e_g is $(-0.945636157362, -0.154606293310, 0.286127859460)$.

All element axes have $\alpha = 90$ degrees from the plane normal in the unconstrained optimum. Element j uses $\theta_j = 360(j-1)/13$ degrees and phase $\phi_j = -\theta_j$ modulo 360 degrees. The reversed winding produces the opposite lateral

direction. The exact element vectors, sky coordinates, delays, and CMB path asymmetries are provided in Section 7 and the machine-readable companion.

The nominal carrier is 100 MHz only under the selected 50-100 MHz implementation constraint. The intrinsic optimization is $\arg \max f_{\eta_m H_{13}}$. At the nominal point, the carrier period is 10 ns, adjacent delay is 0.769230769 ns, and the ideal maximum force is $8.989366088879e-5 M_{\text{active}} f_{H_{13}}$ newtons. The ideal values are 0.898936609 H_{13} N for 100 mg and 0.116861759 H_{13} N for thirteen one-milligram active elements.

The thirteen-element plane is an active source module and is not mechanically quiet by itself. A complete high-frequency force experiment requires antipodal or equivalent carrier-force balancing, chirality modulation, vector force readout, phase and amplitude reconstruction, orientation nulls, and explicit extraction of H_{13} . These requirements convert the geometric result into a falsifiable and quantitatively calibrated experimental program.

Parameter	Final specification
N	13 active linear piezo elements
Coordinate frame	ICRS; local plane basis (e_g, ell)
e_g	(-0.945636157362, -0.154606293310, 0.286127859460)
n_opt	(-0.315865036407, 0.646162683024, -0.694768354089)
ell	(0.077469585422, 0.747375863362, 0.659869519070)
Axis tilt	$\alpha_j = 90 \text{ deg}$
Axis azimuth	$\theta_j = 360(j-1)/13 \text{ deg}$
Positive-force phase	$\phi_j = \phi_0 - \theta_j$
Negative-force phase	$\phi_j = \phi_0 + \theta_j$
Nominal frequency	100 MHz constrained design point
Adjacent delay	0.769230769 ns at 100 MHz
Force law	$F_K = M_{\text{active}} \pi \kappa_F f_{C_{13} H_{13}} (n_{\text{opt}} \times g)$

Appendix A. Complete 27-Node Cell3 Ledger

Address	lambda	Polarity	Kind	x (nm)	y (nm)	z (nm)	Radius (nm)
(-1, -1, -1)	-3	-1	corner	-0.176351350143518	0.118682429904309	0.637311498325592	0.671826810839999
(-1, -1, 0)	-2	1	edge center	-0.181416008956851	-0.076211421277593	0.292444294144131	0.352482076447075
(-1, 0, -1)	-2	1	edge center	-0.158685962221493	0.222342355125059	0.363610955629613	0.454785976633624
(0, -1, -1)	-2	1	edge center	-0.012600729108693	0.091233925961153	0.618567746877440	0.625386652477799
(-1, -1, 1)	-1	-1	corner	-0.186480667770183	-0.271105272459495	-0.052422910037330	0.333198543970607
(-1, 0, 0)	-1	-1	face center	-0.163750621034825	0.027448503943156	0.018743751448153	0.167089839536576
(-1, 1, -1)	-1	-1	corner	-0.141020574299467	0.326002280345807	0.089910412933635	0.366398924016317
(0, -1, 0)	-1	-1	face center	-0.017665387922025	-0.103659925220749	0.273700542695978	0.293205445207418
(0, 0, -1)	-1	-1	face center	0.005064658813332	0.194893851181902	0.344867204181461	0.396159882518847
(1, -1, -1)	-1	-1	corner	0.151149891926133	0.063785422017997	0.599823995429287	0.621855043707172
(-1, 0, 1)	0	1	edge center	-0.168815279848158	-0.167445347238745	-0.326123452733309	0.403600110808719
(-1, 1, 0)	0	1	edge center	-0.146085233112800	0.131108429163905	-0.254956791247826	0.321765879071780
(0, -1, 1)	0	1	edge center	-0.022730046735358	-0.298553776402651	-0.071166661485483	0.307759169052467
(0, 0, 0)	0	1	body center	0.000000000000000	0.000000000000000	0.000000000000000	0.000000000000000
(0, 1, -1)	0	1	edge center	0.022730046735358	0.298553776402651	0.071166661485483	0.307759169052467
(1, -1, 0)	0	1	edge center	0.146085233112800	-0.131108429163905	0.254956791247826	0.321765879071780
(1, 0, -1)	0	1	edge center	0.168815279848158	0.167445347238745	0.326123452733309	0.403600110808719
(-1, 1, 1)	1	-1	corner	-0.151149891926133	-0.063785422017997	-0.599823995429287	0.621855043707172
(0, 0, 1)	1	-1	face center	-0.005064658813332	-0.194893851181902	-0.344867204181461	0.396159882518847
(0, 1, 0)	1	-1	face center	0.017665387922025	0.103659925220749	-0.273700542695978	0.293205445207418
(1, -1, 1)	1	-1	corner	0.141020574299467	-0.326002280345807	-0.089910412933635	0.366398924016317
(1, 0, 0)	1	-1	face center	0.163750621034825	-0.027448503943156	-0.018743751448153	0.167089839536576
(1, 1, -1)	1	-1	corner	0.186480667770183	0.271105272459495	0.052422910037330	0.333198543970607
(0, 1, 1)	2	1	edge center	0.012600729108693	-0.091233925961153	-0.618567746877440	0.625386652477799
(1, 0, 1)	2	1	edge center	0.158685962221493	-0.222342355125059	-0.363610955629613	0.454785976633624
(1, 1, 0)	2	1	edge center	0.181416008956851	0.076211421277593	-0.292444294144131	0.352482076447075
(1, 1, 1)	3	-1	corner	0.176351350143518	-0.118682429904309	-0.637311498325592	0.671826810839999

Appendix B. Plane Membership Ledger

Channel a - bc face family

Plane address	Nine contained node addresses
-1	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (-1, 0, -1); (-1, 0, 0); (-1, 0, 1); (-1, 1, -1); (-1, 1, 0); (-1, 1, 1)
0	(0, -1, -1); (0, -1, 0); (0, -1, 1); (0, 0, -1); (0, 0, 0); (0, 0, 1); (0, 1, -1); (0, 1, 0); (0, 1, 1)
1	(1, -1, -1); (1, -1, 0); (1, -1, 1); (1, 0, -1); (1, 0, 0); (1, 0, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)

Channel b - ac face family

Plane address	Nine contained node addresses
-1	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (0, -1, -1); (0, -1, 0); (0, -1, 1); (1, -1, -1); (1, -1, 0); (1, -1, 1)
0	(-1, 0, -1); (-1, 0, 0); (-1, 0, 1); (0, 0, -1); (0, 0, 0); (0, 0, 1); (1, 0, -1); (1, 0, 0); (1, 0, 1)
1	(-1, 1, -1); (-1, 1, 0); (-1, 1, 1); (0, 1, -1); (0, 1, 0); (0, 1, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)

Channel c - ab face family

Plane address	Nine contained node addresses
-1	(-1, -1, -1); (-1, 0, -1); (-1, 1, -1); (0, -1, -1); (0, 0, -1); (0, 1, -1); (1, -1, -1); (1, 0, -1); (1, 1, -1)
0	(-1, -1, 0); (-1, 0, 0); (-1, 1, 0); (0, -1, 0); (0, 0, 0); (0, 1, 0); (1, -1, 0); (1, 0, 0); (1, 1, 0)
1	(-1, -1, 1); (-1, 0, 1); (-1, 1, 1); (0, -1, 1); (0, 0, 1); (0, 1, 1); (1, -1, 1); (1, 0, 1); (1, 1, 1)

Appendix C. Element Address-Direction Ledger

The vector $2B^{(-1)}d_j$ is the Cell3 half-step address change per metre of physical displacement along the positive axis of element j. Multiplication by scalar displacement gives the local address coordinate. These values permit direct simulation of the array in the supplied W metric.

j	q_a per m	q_b per m	q_c per m
1	-5.580771783559e+09	-1.588530331304e+09	7.343648524313e+08
2	-4.839432380349e+09	-7.262811620392e+08	-7.844109968581e+08
3	-2.989437340284e+09	3.023502688116e+08	-2.123487739945e+09
4	-4.545982321852e+08	1.261716896793e+09	-2.976099032611e+09
5	2.184383852405e+09	1.932039389057e+09	-3.146921902788e+09
6	4.322949921088e+09	2.159754940886e+09	-2.596822889556e+09
7	5.471180260044e+09	1.892696663627e+09	-1.451823047435e+09
8	5.366029136293e+09	1.192044390198e+09	2.577195848845e+07
9	4.031585405080e+09	2.183091126672e+08	1.497462919308e+09
10	1.773554043433e+09	-8.054381516657e+08	2.626103171699e+09
11	-8.907771759203e+08	-1.644669242034e+09	3.153134835428e+09
12	-3.351042079299e+09	-2.107126429465e+09	2.957821307754e+09
13	-5.043623626746e+09	-2.086866345532e+09	2.084906564085e+09

Appendix D. Source Manifest and Data Lineage

The report is generated from the following supplied machine-readable records. Hashes are SHA-256 values of the materialized source bytes. Duplicate library copies were excluded from the unique source set. The figures in this paper were generated from these records and the deterministic equations stated in the text.

Role	Filename	Bytes	SHA-256
single_cell	K-Field_single_cell3_lateral_force_reduction_1784985105.json	47891	62e4472aacc7df503cc4b5120d7abce97589891e8d22843177ee547e35b71a6f
oscillator	K-Field_circularly_polarized_elastic_oscillator_1784991135.json	137270	3c4f03389b05a3114cc13c7c2d6d5a468371fc8dbdfced274b9ac64455310483
linear_inertia	K-Field_linear_inertia_derivation_1784989727.json	42560	4651a532fe43e8091748ef4f7e28bb899e1a2ea798d689ba1724e99530cd55ab
fourier	K-Field_fourier_geomery_hidden_architecture_1784910918(1).json	24374	1e55b6ddff6ca08a51cf31ea8663e9ff383fd1c648695d0cb6802a8dce38b3d3
materials	K-Field_materials_for_coherently_controlled_angular_momentum_density_1784992238_machine_readable.json	18071	96c84b9d8056b8bb3f335c9b6239340dc1380a2c099d749f473c60868d304ef1
noise_immunity	K-Field_functionally_immune_to_non-coherent_noise_in_addresses_determination_1784513477(1).json	50933	5d877d4899b06cf95a57528380376b01f7159077662c817670778f549ef63569
spectral_fabric	K-Field_self-indexing_spectral_fabric_1784510144(1).json	17035	5c8d9147b4f0e149716122f23620380ecbaef92d6fb6510e97a3c7c657609726

Appendix E. References

[1] P. Dwayne Esterline, K-Field Single-Cell3 Lateral Force Reduction: Audited Cell3 Coordinate Reduction, True-Scale Geometry, and Phase-Ordered Recovery of Tubular-Rotor Lateral Acceleration, corrected machine-readable edition, epoch 1784985105, Krytronx, LLC.

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